

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_132841 | Base: main-v1.0

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Decision: GO_WITH_CONSTRAINTS | Confidence: 78%

CHANGE REQUEST

modifica eliminando il pannello solare di destra

CASCADE ISSUES IDENTIFIED

[MAJOR] ISSUE-001 — Power margin reduction

Subsystem: solar | Req: L1-PWR-001 | Type: constraint_violation

Removing the right solar panel reduces power generation capacity, potentially exceeding peak power consumption limits during imaging mode.

[MINOR] ISSUE-002 — Mass reduction

Subsystem: structure | Req: None | Type: reanalysis_required

Removing the right solar panel reduces total payload mass, potentially affecting overall structural dynamics and compliance with mechanical requirements.

[MINOR] ISSUE-003 — Thermal balance impact

Subsystem: thermal | Req: None | Type: reanalysis_required

Removing the right solar panel may alter thermal distribution and affect temperature stability across orbital cycles, potentially impacting thermal requirements.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-31C §5.4 — Thermal balance test to validate TMM with modified solar panel configuration

Reason: Removing the right solar panel affects thermal distribution and requires revalidation of thermal models.

ECSS-E-ST-32C Rev.1 §9 — Shock test per launcher SRS ICD with modified payload configuration

Reason: Removing the right solar panel changes structural dynamics and requires revalidation of shock response.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	61%	WARN	ΔT 14.8°C — margin < 5%	—
mechanical	88%	OK	1st mode 112 Hz > 80 Hz req	—
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

- L1-PWR-001 — verification evidence invalidated, re-analysis required
- L1-MAS-001 — verification evidence invalidated, re-analysis required